

Ethanol Blending in Gasoline – Romania

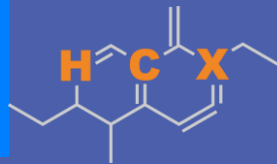
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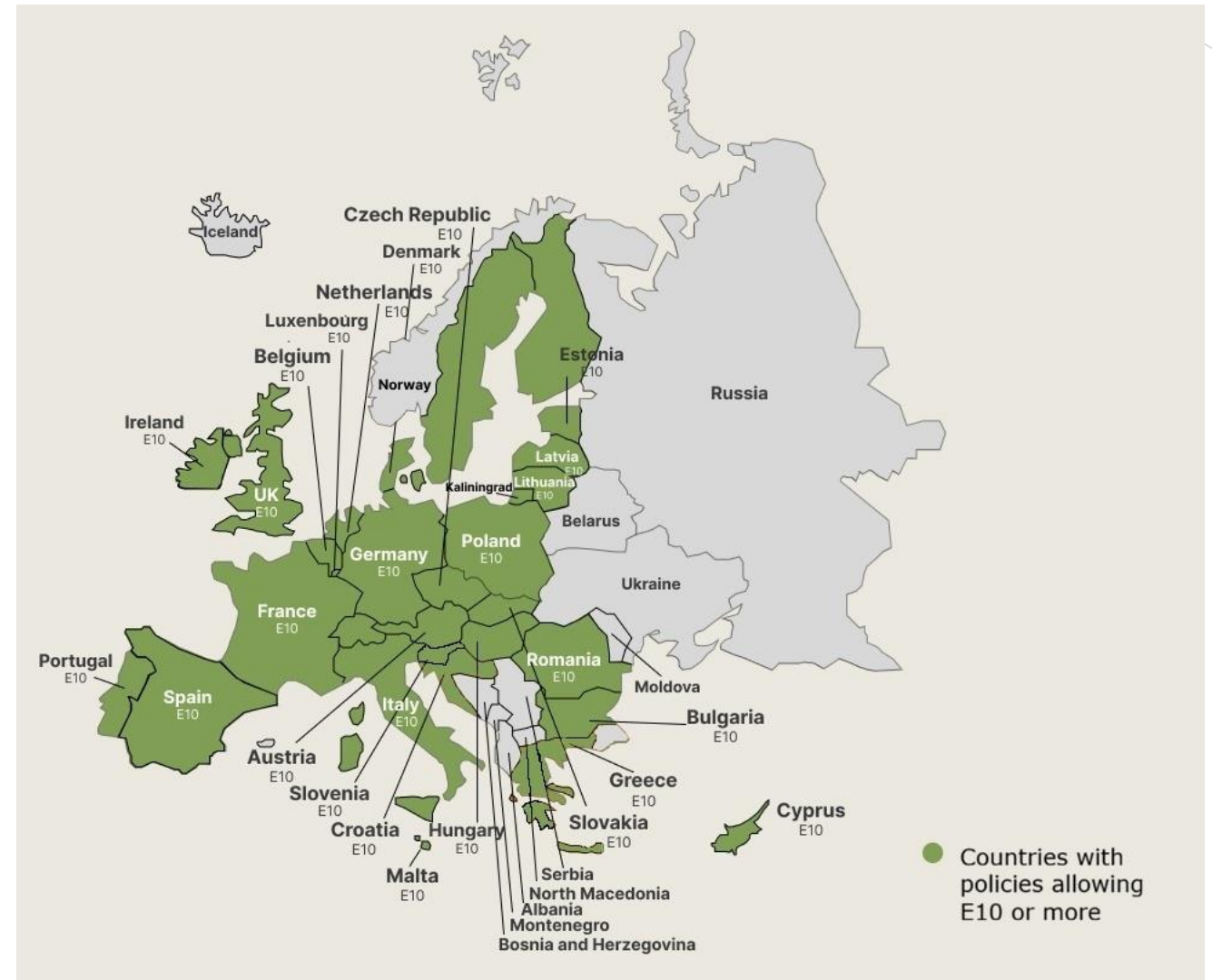


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Ethanol Blending in Europe

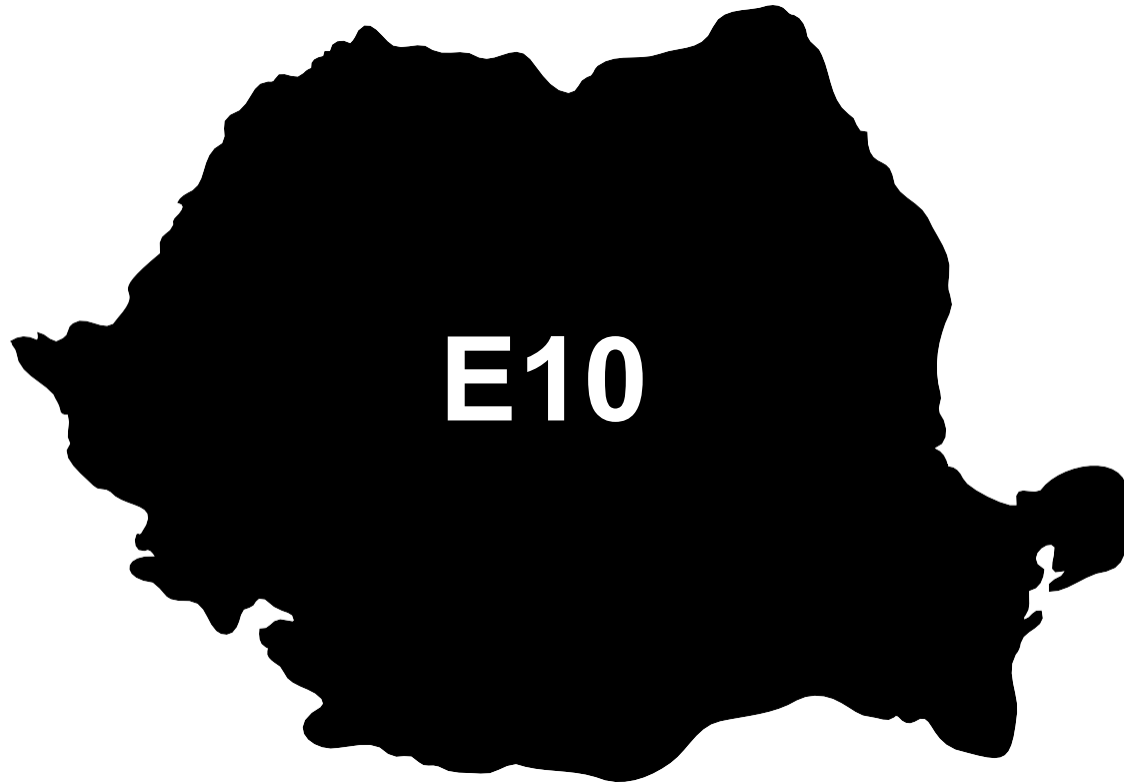
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Romania

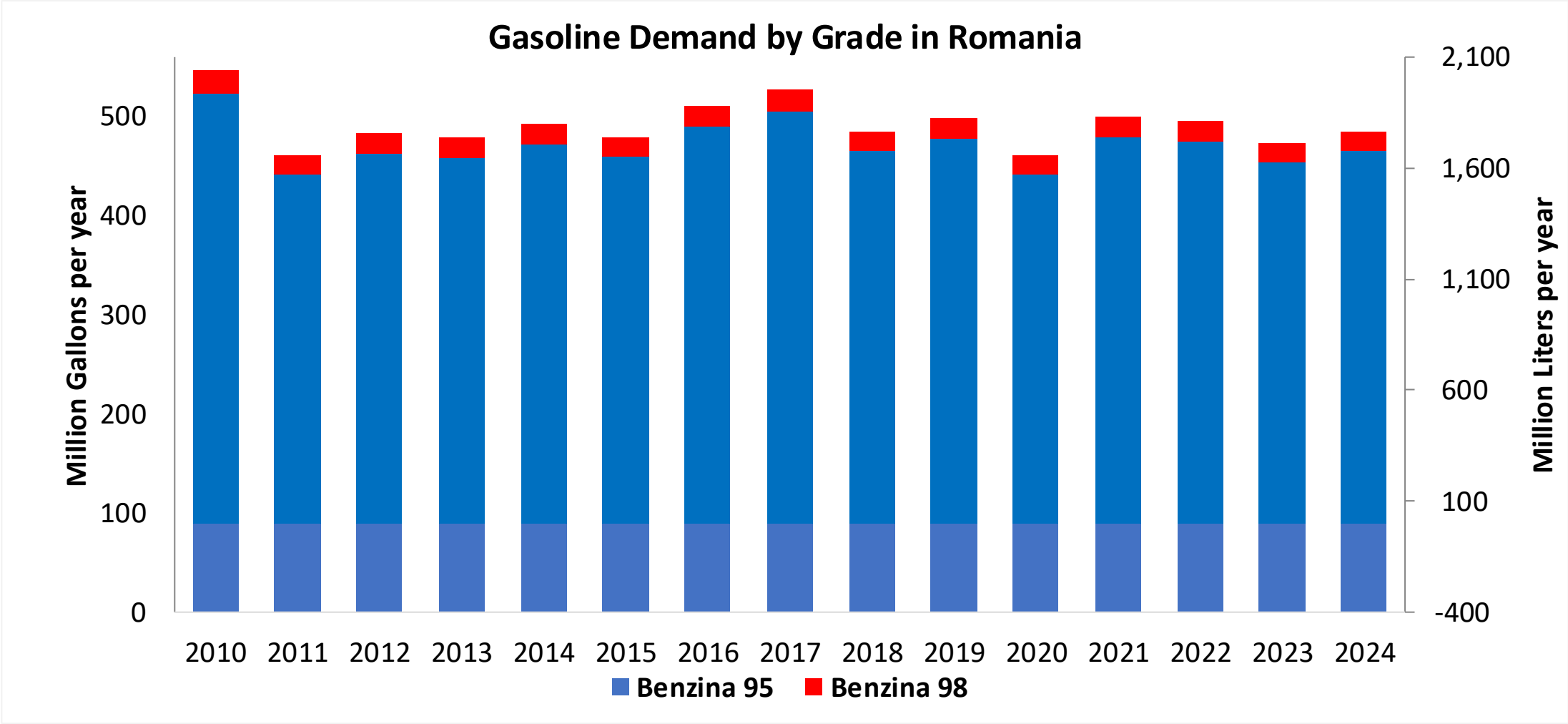


In 2024, gasoline consumption in Romania was 1,762 million liters (465 million gallons) and growth rate was -0.7%. Gasoline production and net imports were 4,358 and -2,596 million liters (1,151 and -686 million gallons) correspondingly. Romania complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.2 RON.

In 2024, ethanol consumption in Romania was 144 million liters (38 million gallons) representing 8.2% of gasoline demand. Ethanol production and Net Imports were 107 and 37 million liters (38 and 10 million gallons) correspondingly. Ethanol sources are Corn 28% and Cereals 72%.

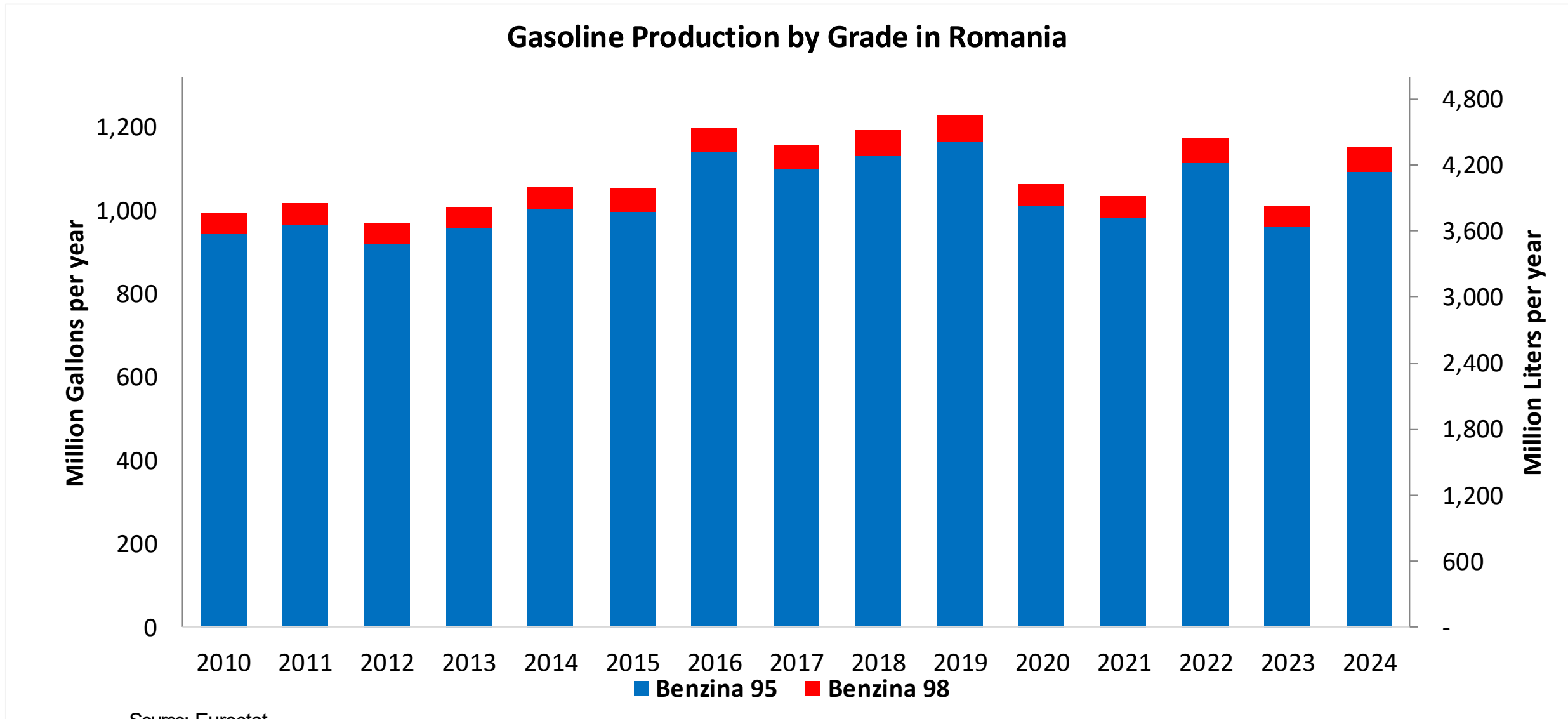
Source: Eurostat, European Commission

Gasoline Demand in Romania



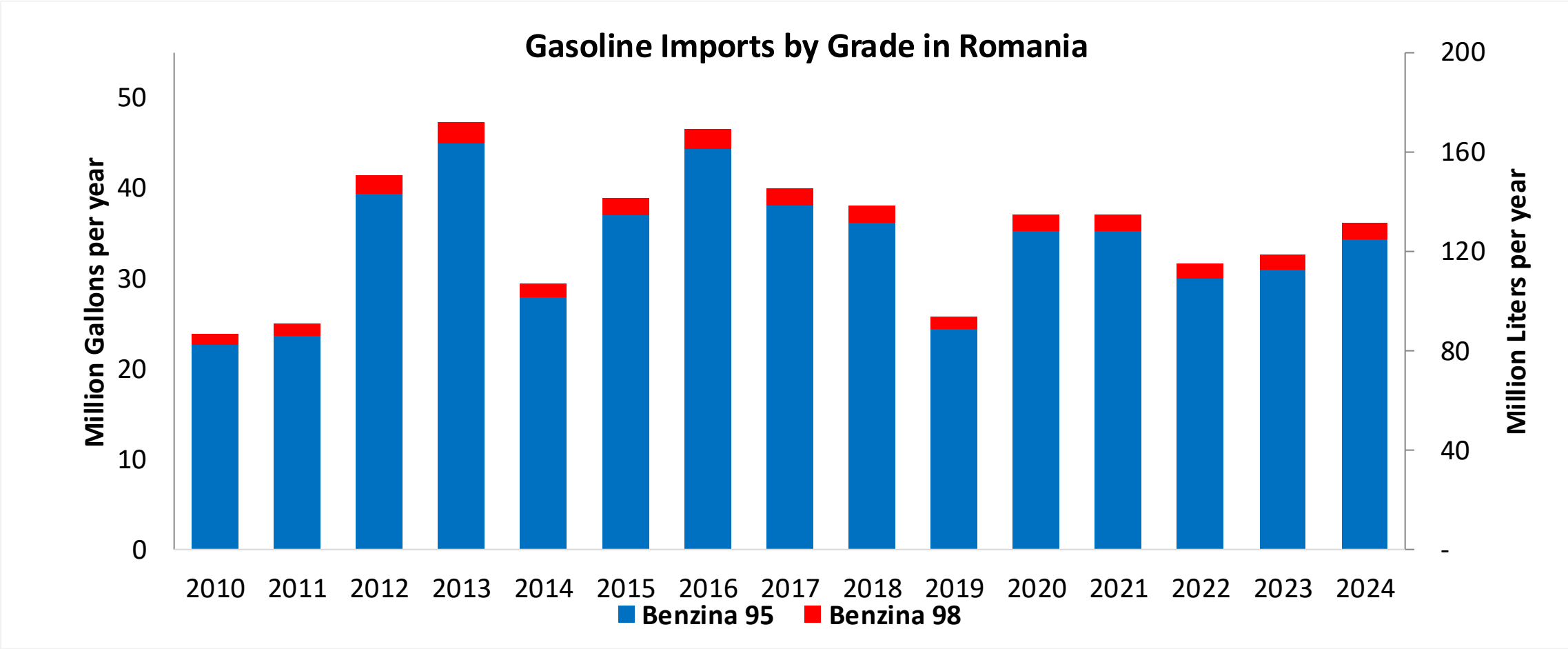
Source: Eurostat

The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexadiene. The structure features a six-membered ring with two double bonds. Substituents include a hydrogen atom (H), a carbon atom (C), and a group labeled 'X' (which is a vinyl group, -CH=CH₂). There are also methyl groups attached to the ring.



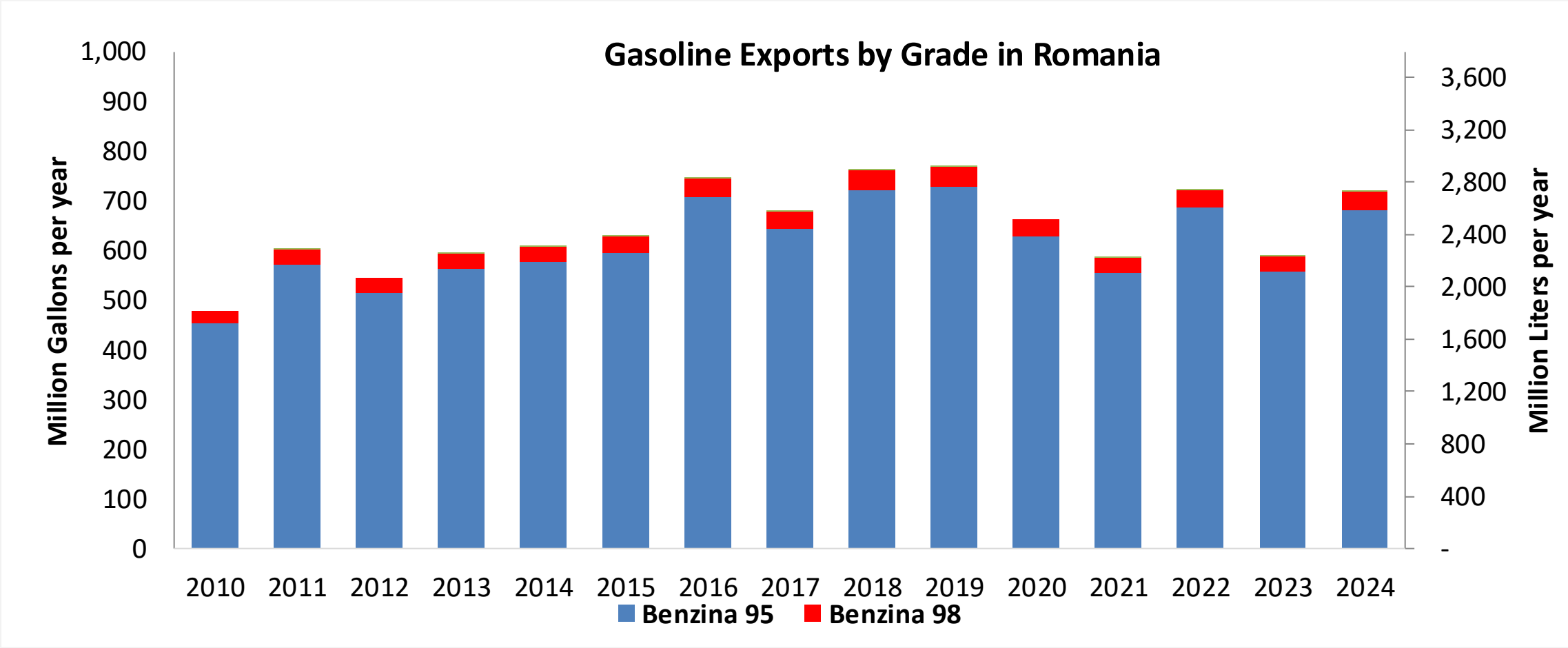
Source: Eurostat

Gasoline Imports in Romania



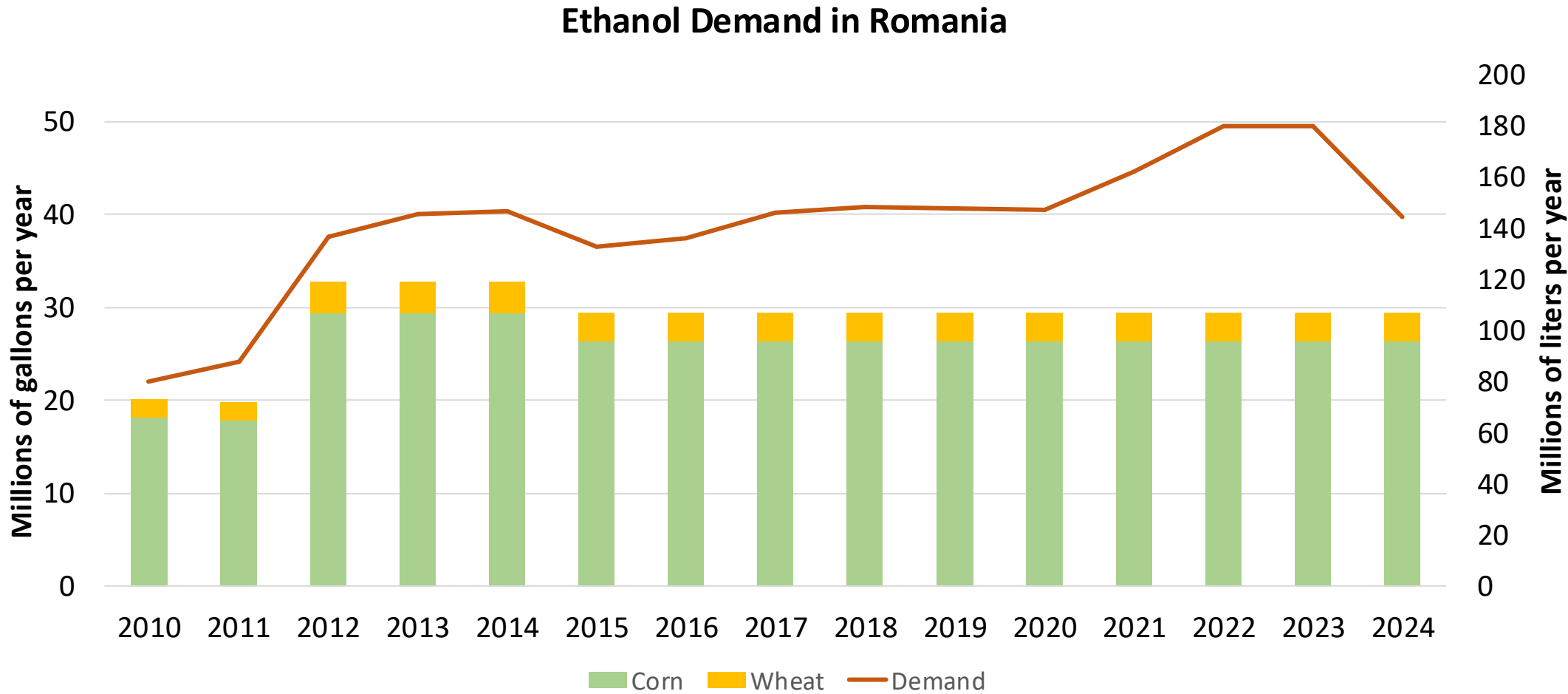
Source: Eurostat

Gasoline Exports in Romania



Source: Eurostat

Ethanol Balance in Romania



Source: Eurostat

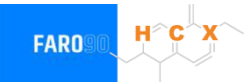
Gasoline Quality in Romania

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	10.0							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

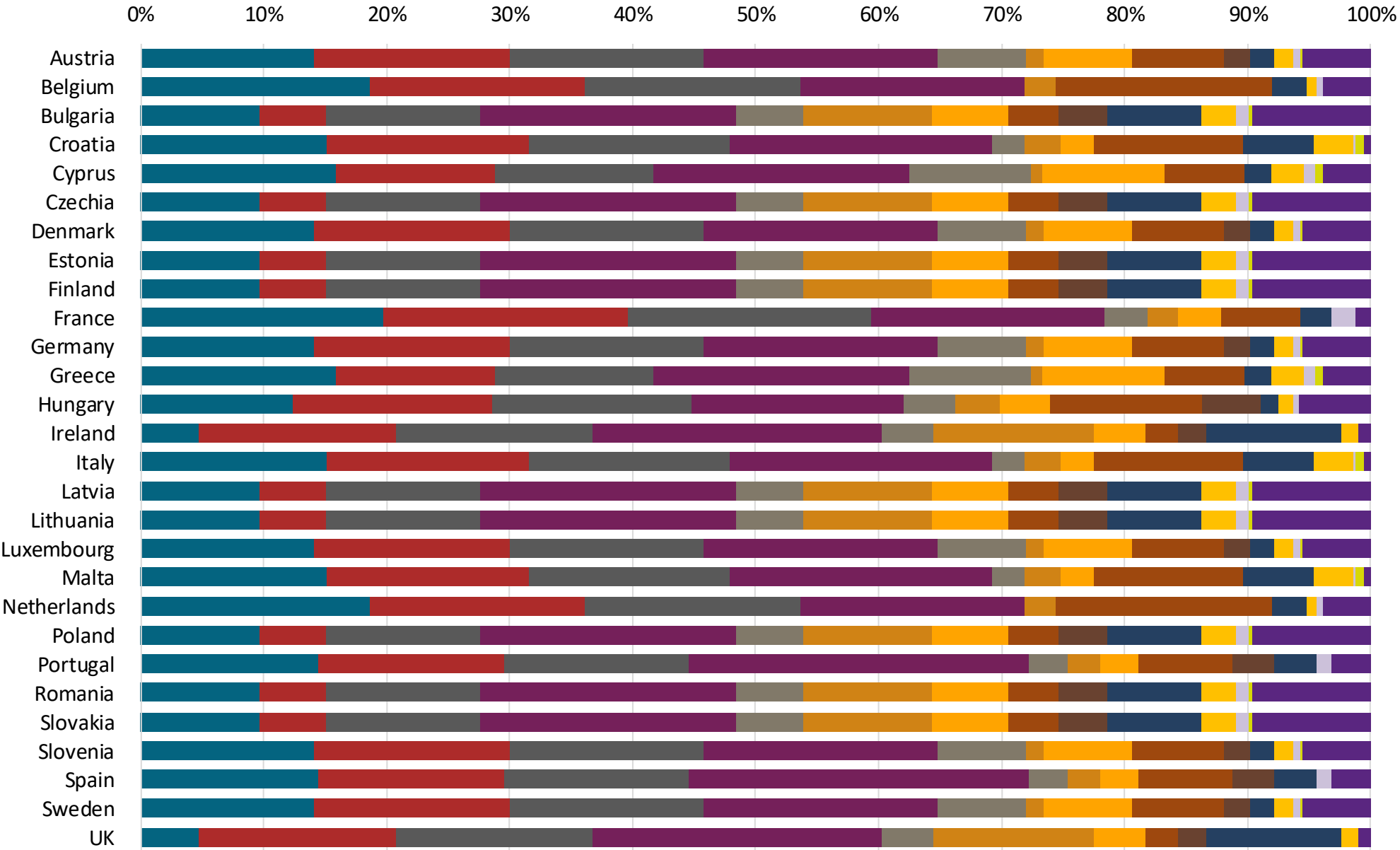
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source: Faro90



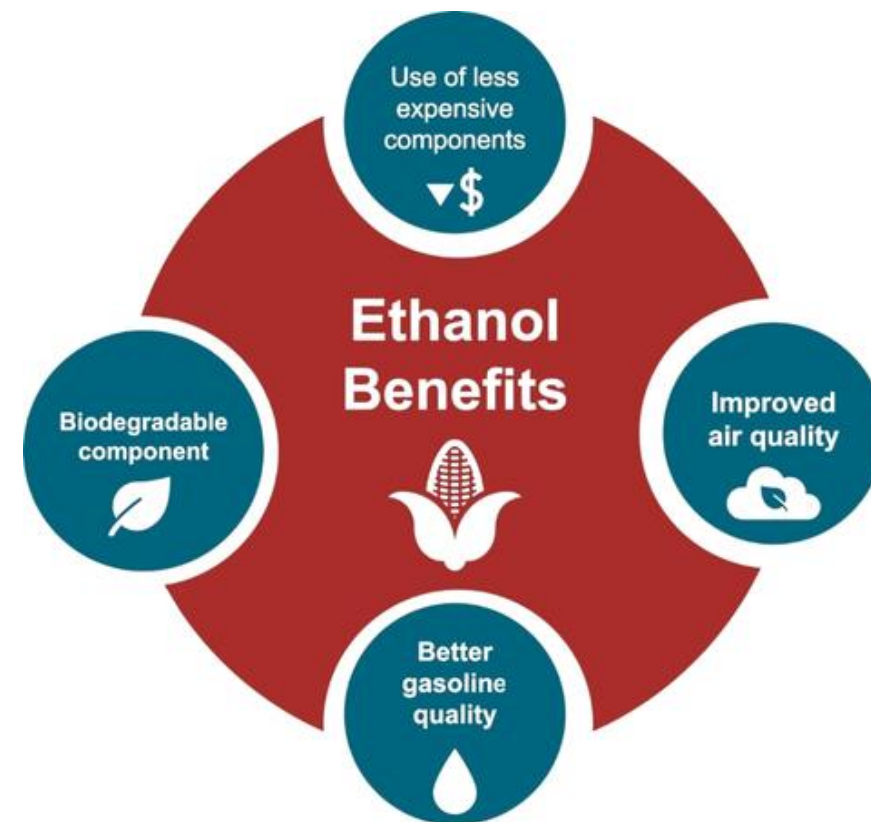
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

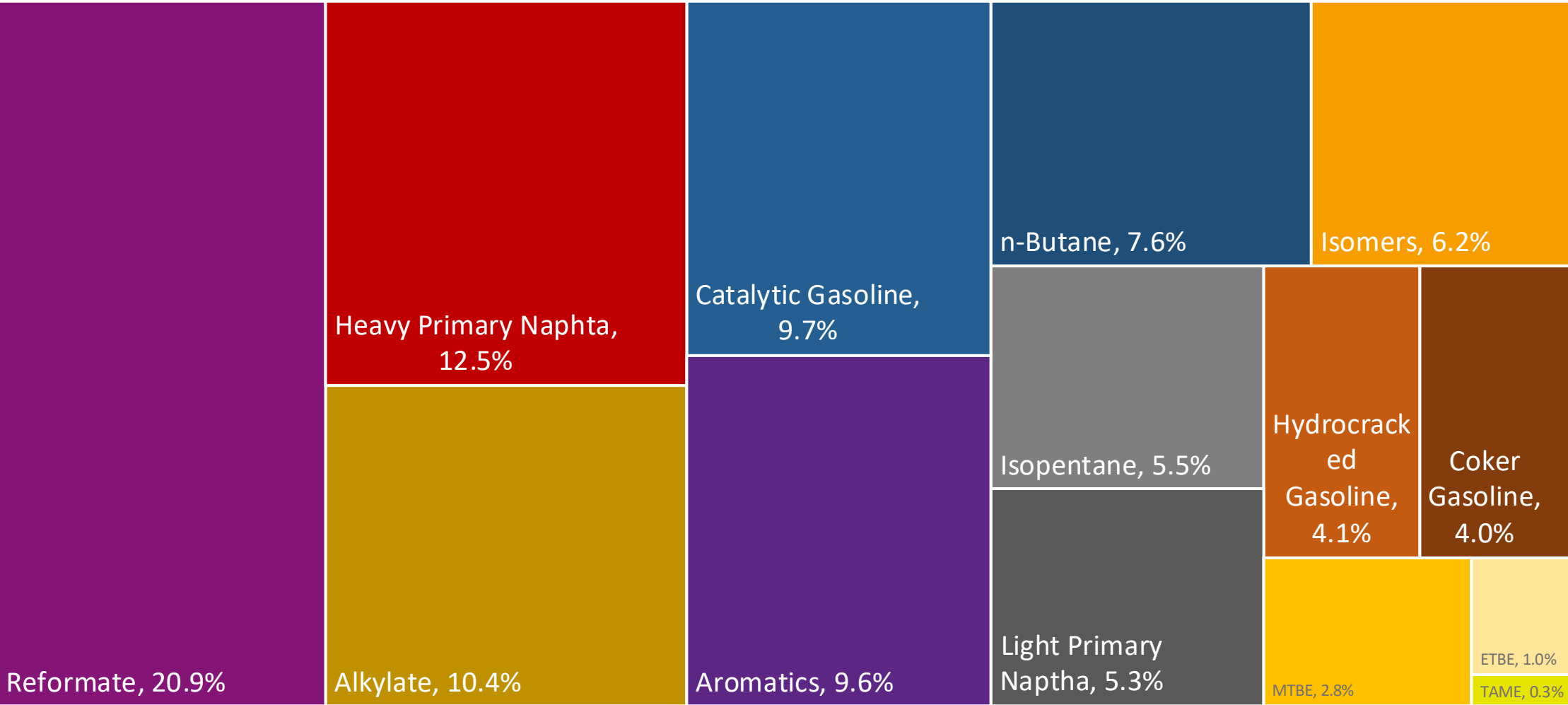
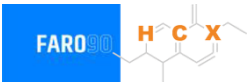
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

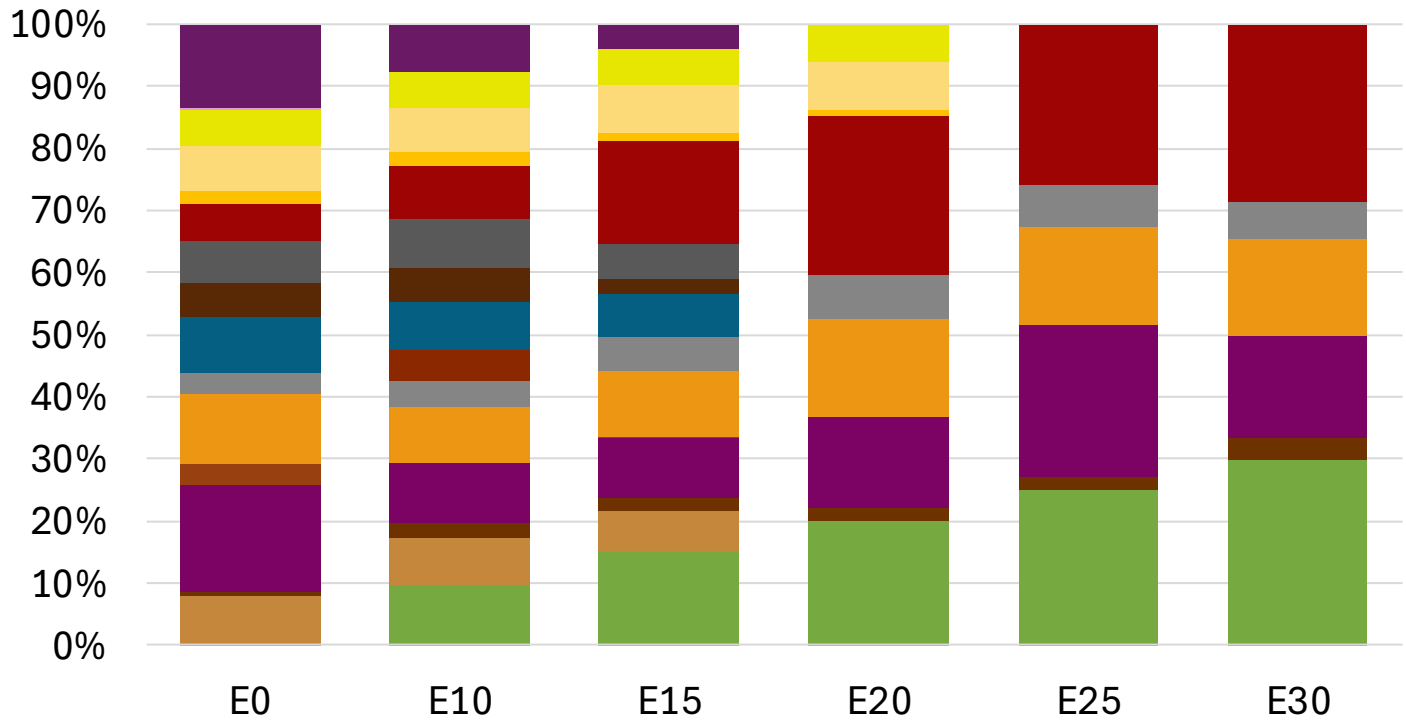
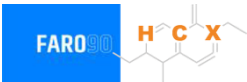
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Romania Available Blending Components



Romania – Gasoline 95 – Constant Octane

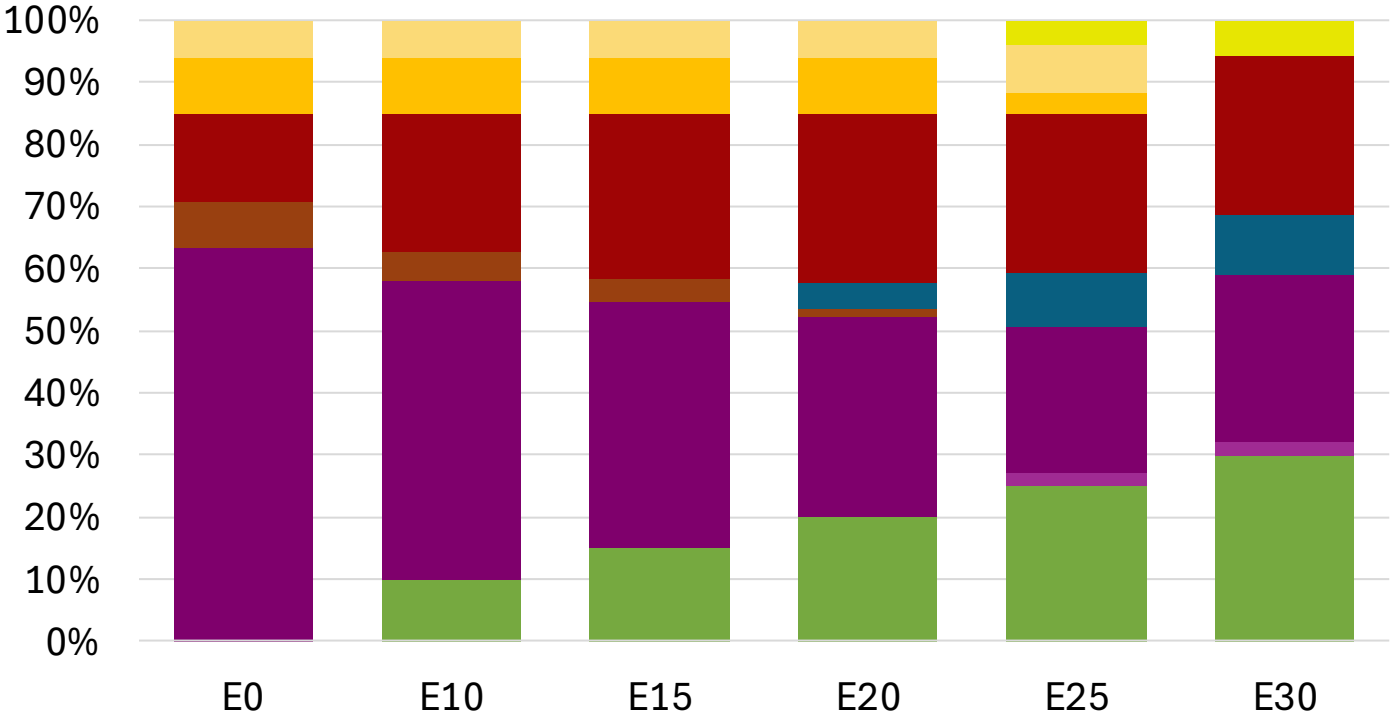
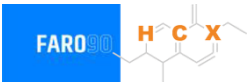


Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.284	\$ 2.146	\$ 2.046	\$ 1.931	\$ 1.885	\$ 1.832

Source: Faro90

Romania - Gasoline 98 - Constant Octane

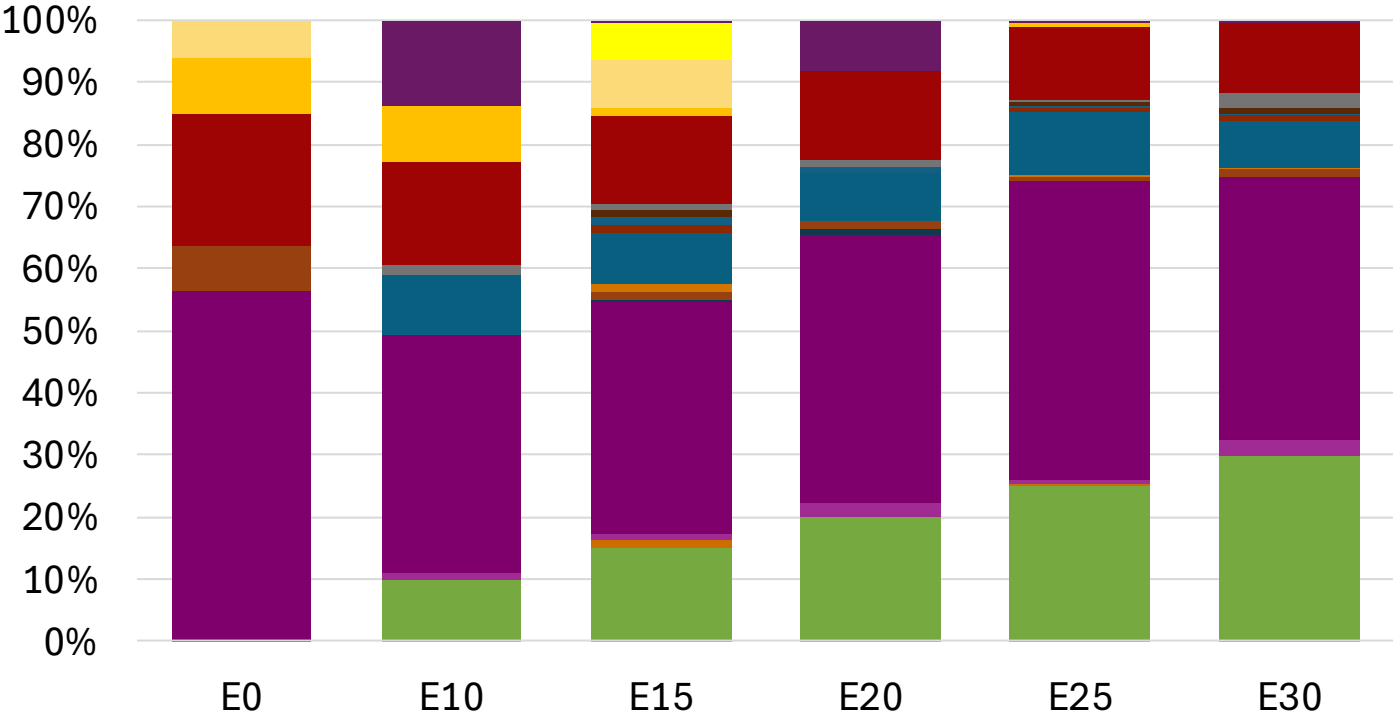
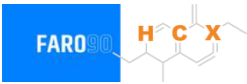


Average CIF 2024
Prices

Octane (RON)	98.0					
Price (US\$/gal)	\$ 2.304	\$ 2.187	\$ 2.118	\$ 2.046	\$ 1.981	\$ 1.931

Source: Faro90

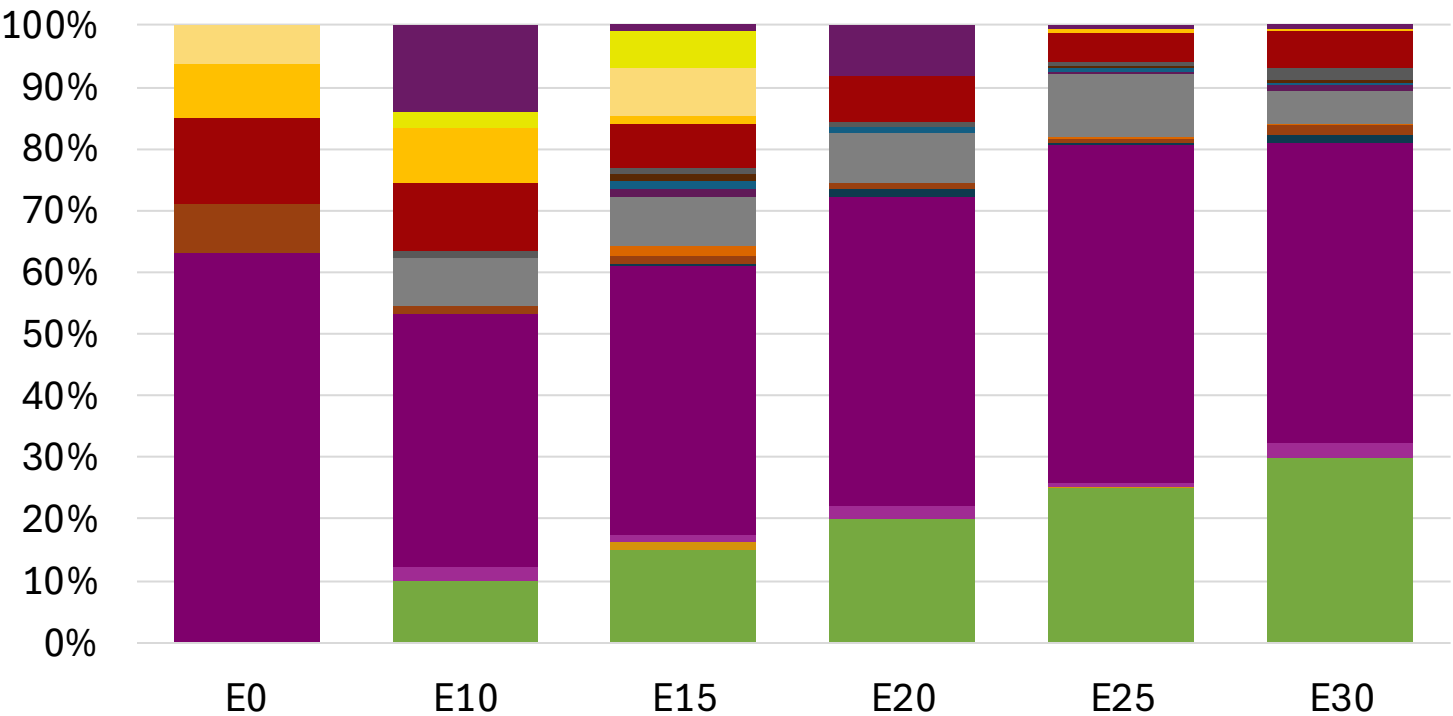
Romania – Gasoline 95 – Increased Octane



Average CIF 2024
Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 2.172	\$ 2.172	\$ 2.172	\$ 2.172	\$ 2.172	\$ 2.172

Romania – Gasoline 98 – Increased Octane

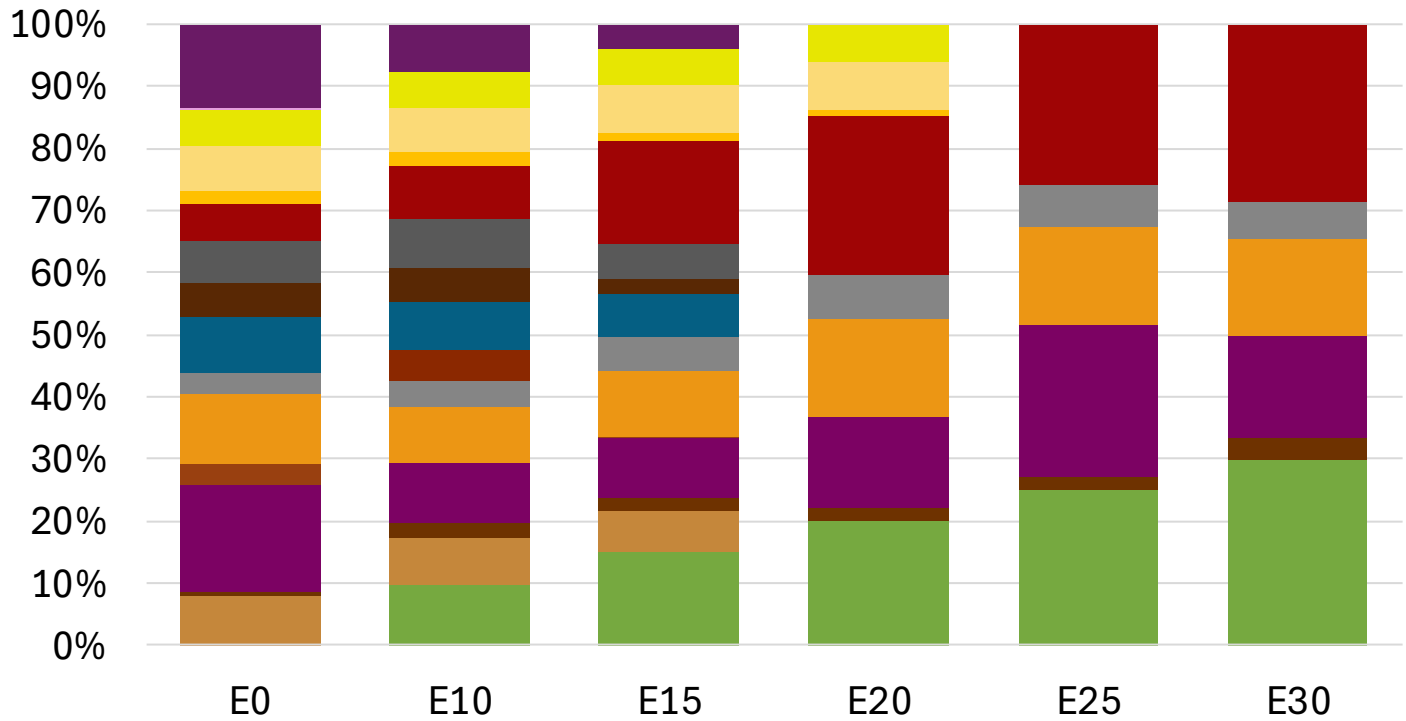
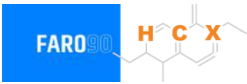


Average CIF 2024
Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.304	\$ 2.304	\$ 2.304	\$ 2.304	\$ 2.304	\$ 2.304

Source: Faro90

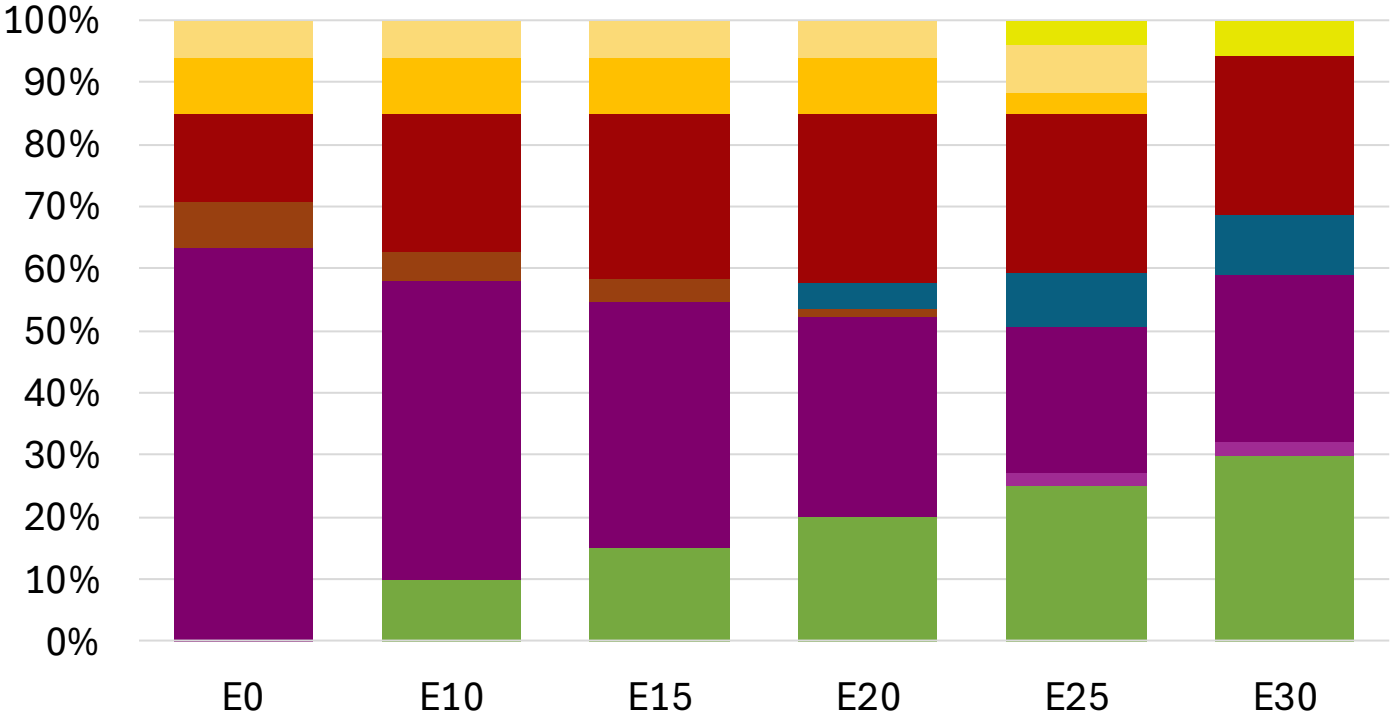
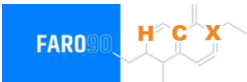
Romania – Gasoline 95 – Constant Octane



Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.274	\$ 2.136	\$ 2.036	\$ 1.921	\$ 1.875	\$ 1.822

Romania - Gasoline 98 - Constant Octane

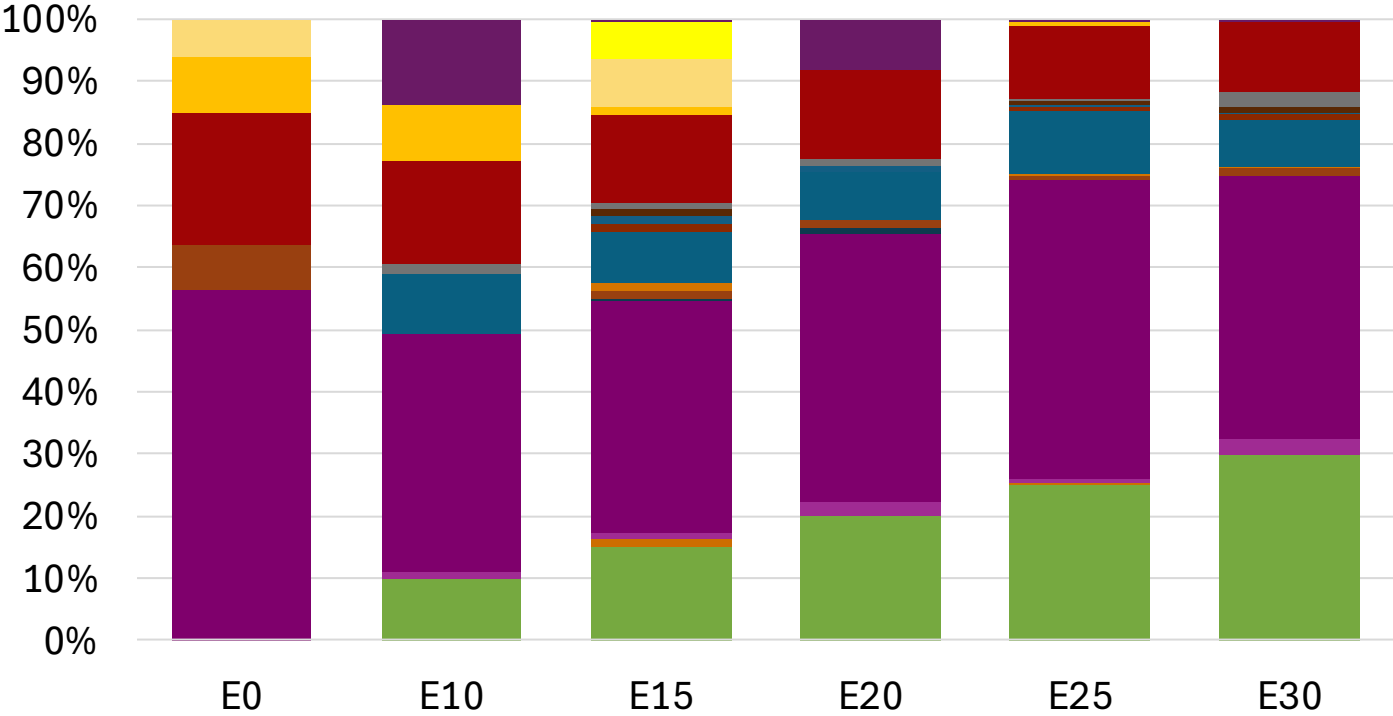


Average CIF 2024
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.294	\$ 2.177	\$ 2.108	\$ 2.036	\$ 1.971	\$ 1.921

Source: Faro90

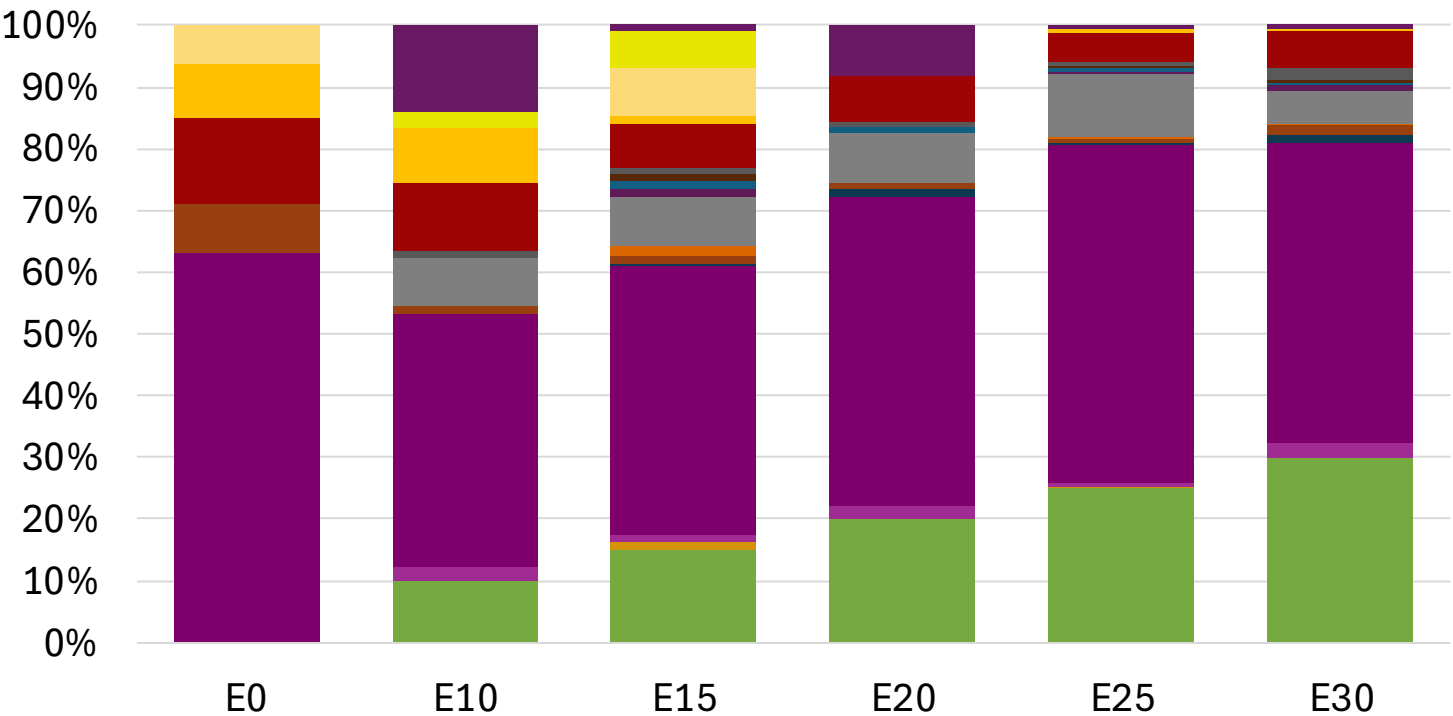
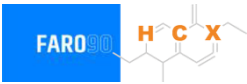
Romania – Gasoline 95 – Increased Octane



Average CIF 2024
Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162

Romania – Gasoline 98 – Increased Octane



Average CIF 2024
Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294	\$ 2.294

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

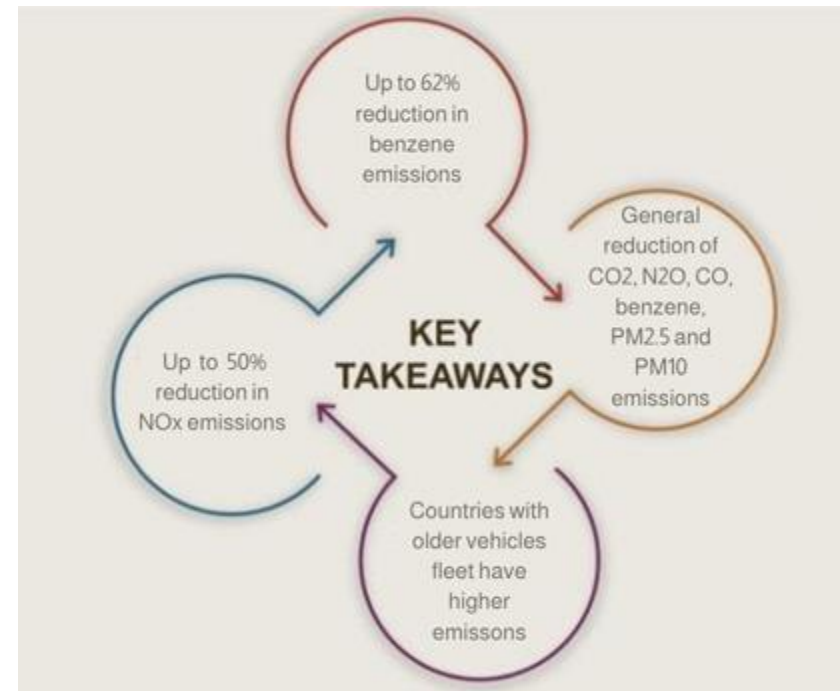
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

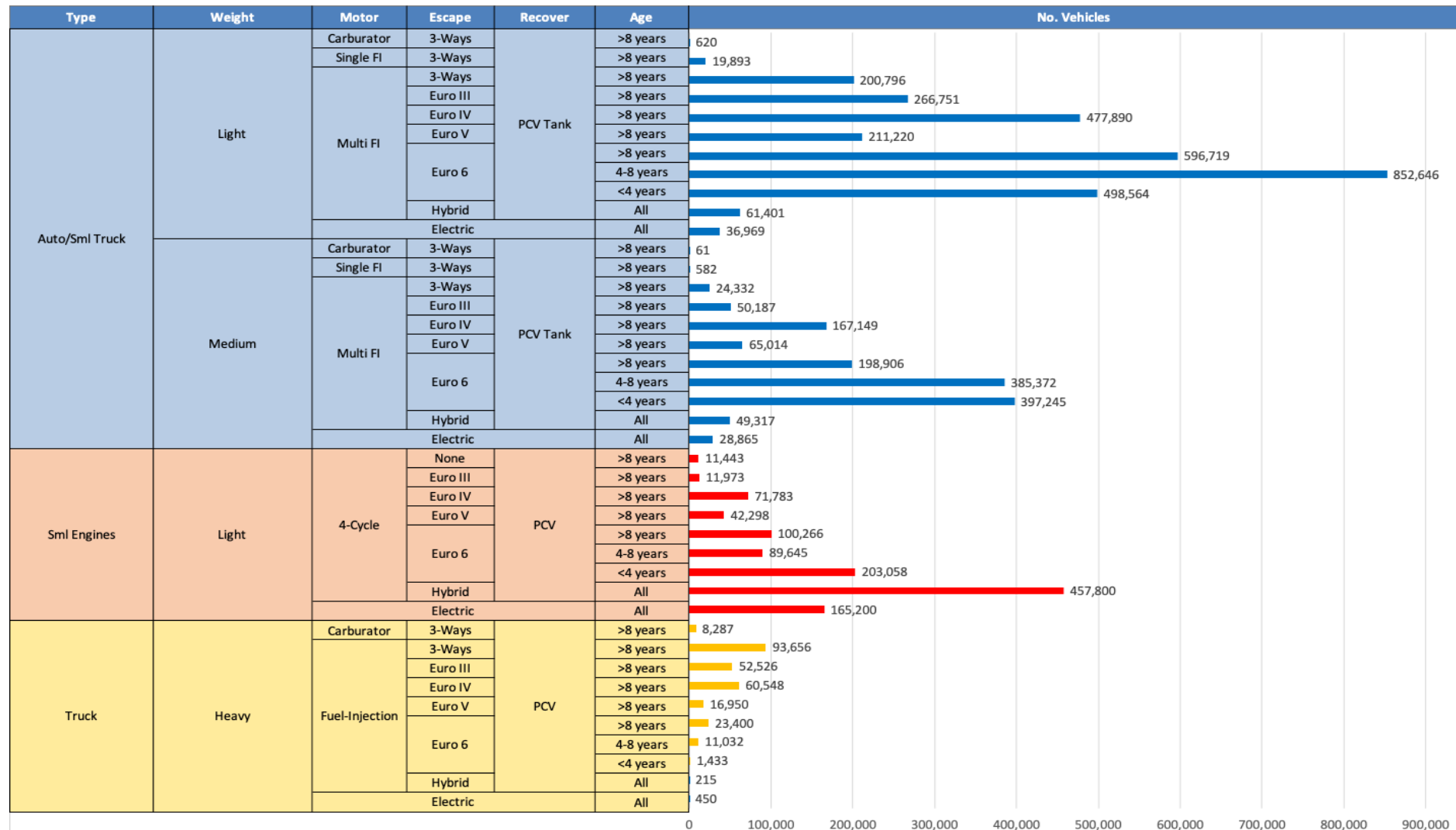
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



Gasoline Vehicle Fleet - Romania

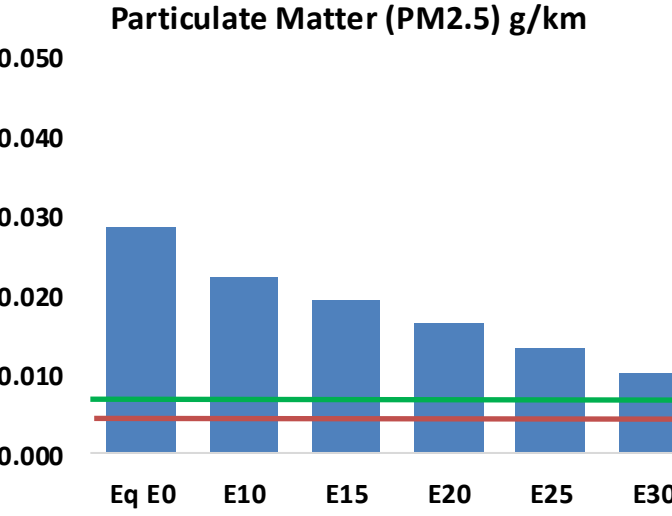
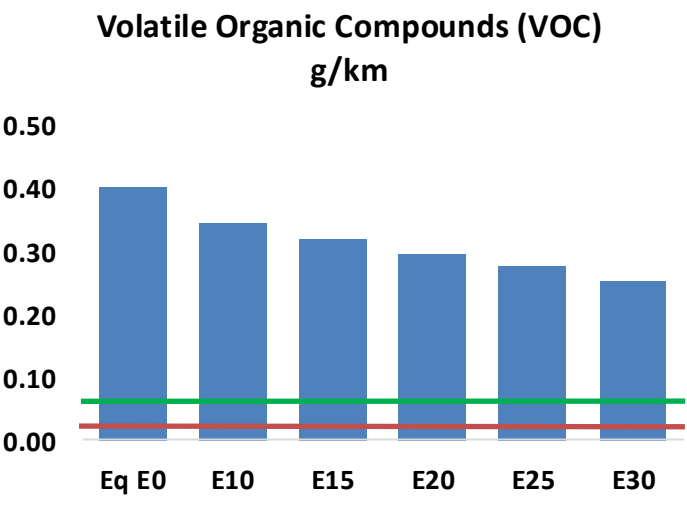
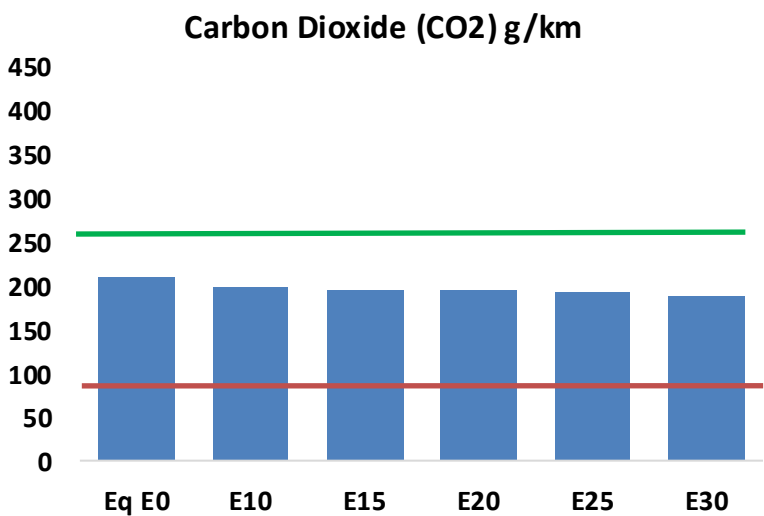
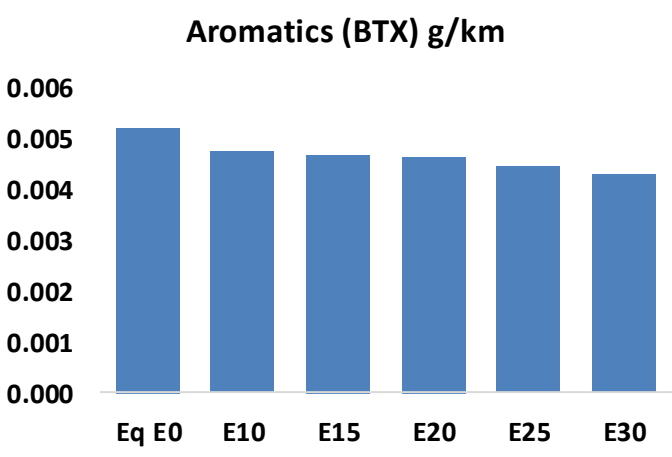
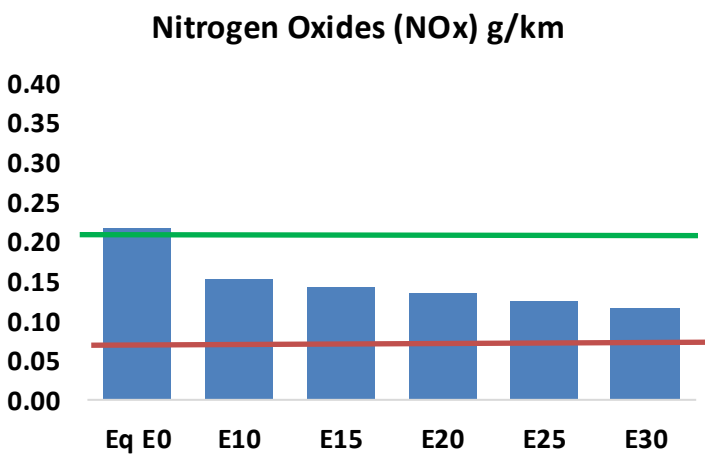
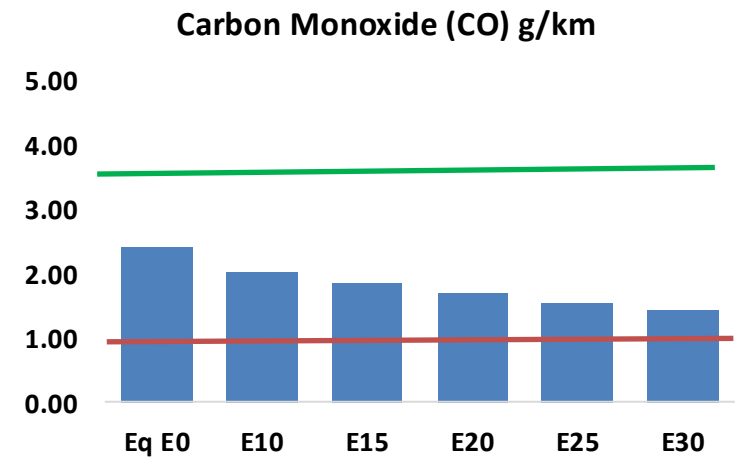
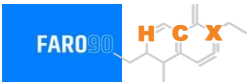


Vehicle Fleet: 6,012,464
Gasoline Vehicle Fleet: 2,699,932

Source: Eurostat, ACEA

Romania – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Romania – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	171,877	157,764	143,651	131,911	120,016	110,845	101,081	-16%	-30%
CO2	14,864,151	14,481,930	14,120,943	13,837,038	13,696,551	13,615,510	13,364,537	-5%	-8%
VOC	28,470	26,412	24,354	22,693	21,033	19,760	18,049	-14%	-26%
NOx	15,404	11,553	10,783	10,163	9,584	8,947	8,272	-30%	-38%
SOx	169	157	144	133	123	111	100	-15%	-28%
NH3	5,167	5,116	5,065	5,089	5,146	5,186	5,220	-2%	0%
N2O	247	246	245	246	251	254	257	-1%	2%
CH4	6,222	6,222	6,222	6,315	6,452	6,570	6,682	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	175	168	161	156	152	149	144	-8%	-14%
Acetaldehyde	489	561	823	1,253	1,707	1,950	2,305	68%	249%
Formaldehyde	1,766	1,717	2,002	2,350	2,462	2,724	2,965	13%	39%
Benzene	372	358	339	333	331	316	306	-9%	-11%
PM2.5	921	818	715	621	527	428	324	-22%	-43%
PM10	1,108	984	860	747	634	515	390	-22%	-43%

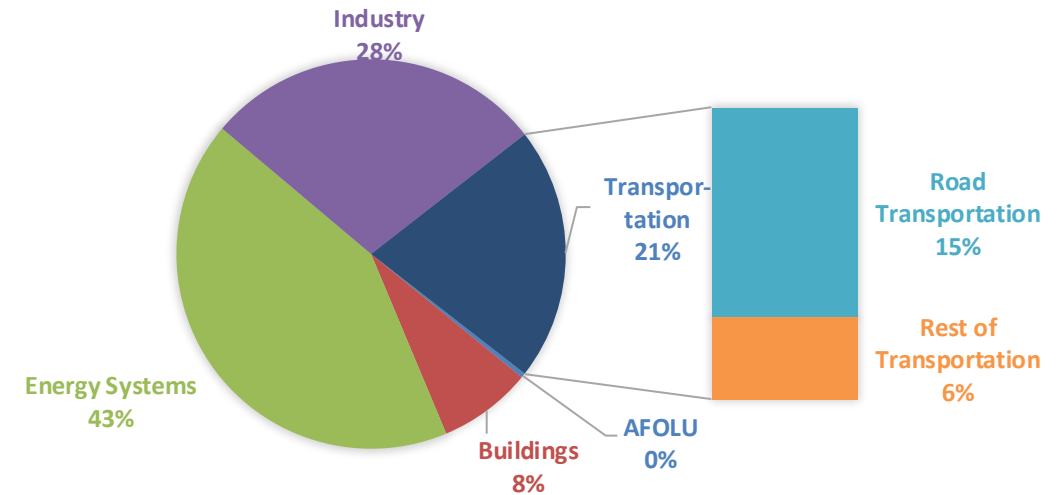
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

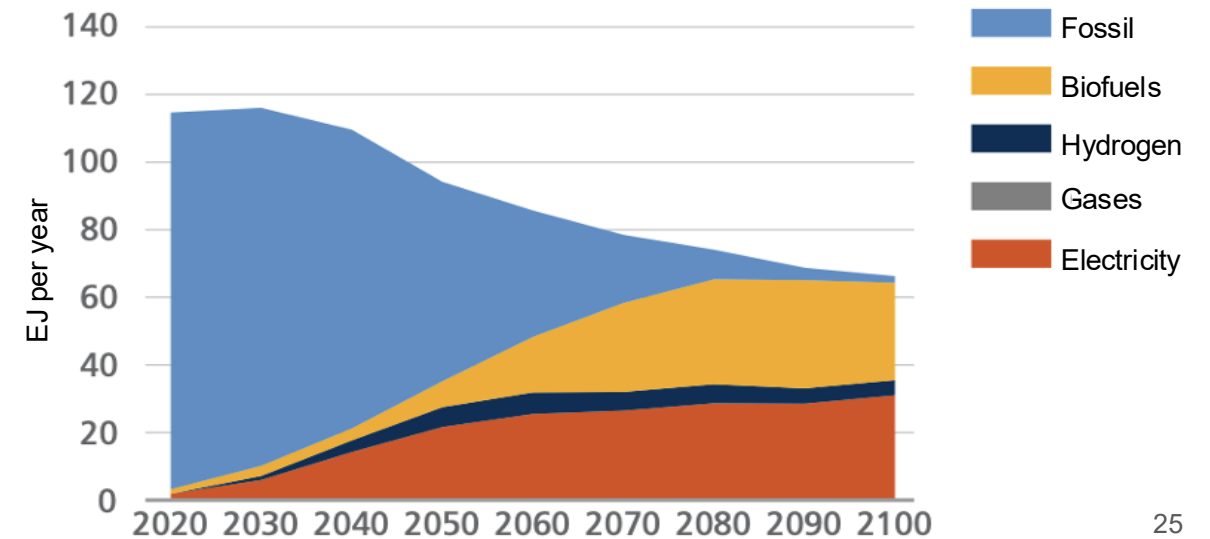
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

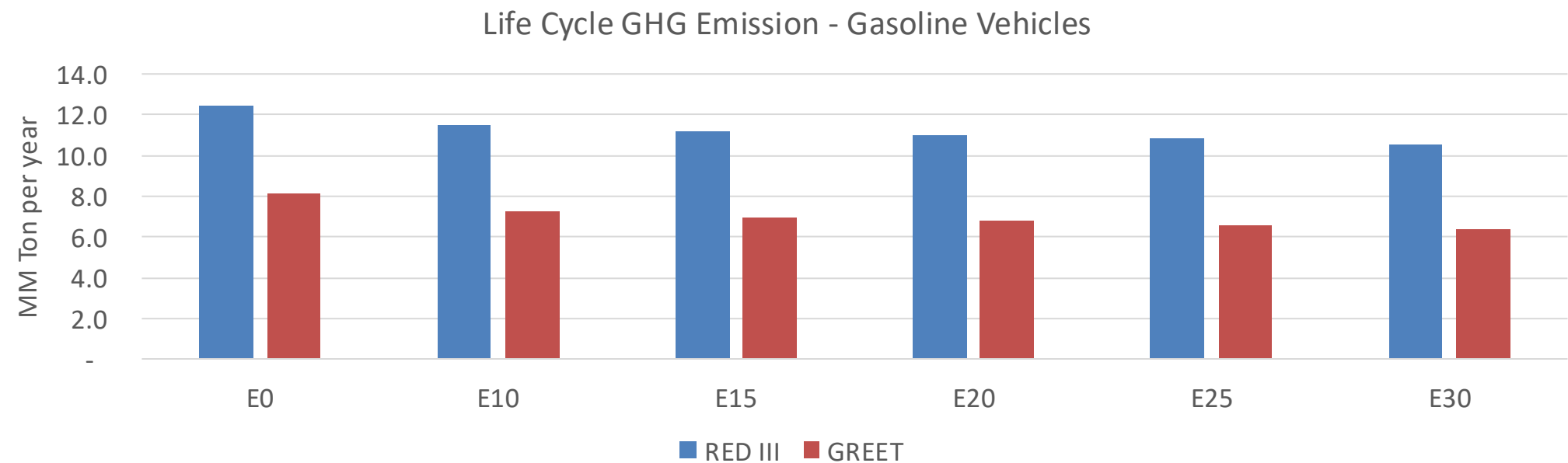
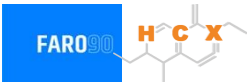
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Romania – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	76.7
Target @ 2035 (MMT/year)	43.2
Delta	33.5

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.3%	14.4%	16.7%	18.9%	22.0%
RED II/III	0.0%	7.8%	10.0%	11.7%	13.3%	15.5%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.7%	3.5%	4.1%	4.6%	5.3%
RED II/III	0.0%	2.9%	3.7%	4.4%	5.0%	5.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90